

DEREK WANG

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Education

University of Waterloo

Bachelor of Applied Science in Computer Engineering

Waterloo, ON, Canada

Expected Apr 2030

Experience

Autonomy Team Member

Waterloo Aerial Robotics Group

Aug 2025 – Present

Waterloo, ON, Canada

- Competing in the 2026 National Annual UAS Student Competition with focus on autonomous flight systems
- Implemented computer vision algorithms using YOLOv11 and OpenCV to detect landing pad zones for autonomous drone navigation and precision landing capabilities

Robotics Software Programmer – VEX Robotics Competition

Checkmate Robotics Club

Sept 2024 – May 2025

Markham, Ontario

- Developed software solutions for Team 16868C Rushdown in the VEX Robotics Competition, earning invitation to the 2025 VRC World Championships
- Implemented advanced robotics systems using Okapi PROS Library, Odometry positioning, and PID Controllers, optimizing robot performance for precise motion control and error handling
- Designed and refactored autonomous navigation algorithms, creating 15-second match routes and 60-second programming skills routines that achieved 5th place out of 80 teams at Provincial Championships

Youth Badminton Instructor

City of Richmond Hill

Mar 2024 – Aug 2025

Richmond Hill, Ontario

- Analyzed student skill levels and developed personalized coaching strategies for 14 beginner to intermediate students (ages 9–16), resulting in improved technique and game performance across 8-week terms
- Designed and implemented a comprehensive badminton curriculum focused on real game situations, which was adopted by all badminton classes at Rouge Woods and Richvale Community Center, improving student engagement
- Built individual mentorship programs through personalized feedback delivery, optimizing lesson plans to address specific performance gaps and foster an inclusive learning environment

Robot Designer and Builder – FIRST Tech Challenge

Dr. X Robotics Academy

July 2019 – May 2022

Richmond Hill, Ontario

- Architected hardware solutions for Team 16417 Dr. X Academy in the FIRST Tech Challenge, earning invitation to the 2022 FTC World Championships
- Designed and built 8 iterations of the team's robot, developing specialized components including drivetrain, arm, claw, turret systems and a backup robot for competition reliability
- Implemented autonomous control systems using Java, creating 30-second match routines with Dead Reckoning position estimation for precise navigation

Projects

Fruit Ninja AI | Python | YOLOv11 | [Source Code](#)

Aug 2025 – Sept 2025

- Architected and implemented an automated computer vision pipeline using YOLOv11 object detection,
- Built comprehensive dataset of 816 images (514 training, 302 testing), optimizing model performance through strategic data augmentation and validation techniques

TTC Escape | C# | Unity | Git | [Source Code](#)

July 2025 – Aug 2025

- Designed innovative puzzle game concept for the 2025 GMTK Game Jam, leading a team of 3 developers
- Developed and implemented core gameplay mechanics using Unity and C#, creating a Toronto Transit Commission-inspired puzzle game featuring looping mechanics that enhanced player engagement

Skills

Languages: Java, C++, Python, HTML5, CSS, JavaScript, C#, TypeScript

Engineering Tools: Git, Fusion 360, Unity

Technologies: ReactJS, OpenCV, YOLOv11, Tailwind CSS